

SATHISH PALANIVELU RAJMOHAN

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CAREER PROFILE

Software Engineering student at UNSW with hands-on experience across full-stack automation, cloud-native architectures, and data engineering. Shortlisted for the Macquarie Industry Prize for implementing a serverless logistics platform. Industry and technical background includes developing custom algorithms to automate workflows at PCL Construction, building a predictive risk intelligence platform for global supply chains, and engineering deep learning systems for medical diagnostics. Experienced in using Python, AWS, and SQL to build scalable backends and data pipelines, transforming complex requirements into reliable, production-ready solutions.

EDUCATION

Bachelor of Engineering (Honours) (Software Engineering) September 2024 – December 2026
University of New South Wales, Sydney

Bachelor of Computer Science & Engineering September 2022 – May 2024
Manipal Academy of Higher Education, India

EXPERIENCE

Software Engineer Intern May 2026 – Present
Rhombus AI, Sydney

- Engineering automated test suites for an AI-powered data pipeline platform, designing test cases covering expected and edge-case model behaviours to ensure consistent output quality across releases.
- Developing structured evaluation frameworks to benchmark AI model performance, systematically identifying failure modes, hallucinations, and quality regressions to inform model iteration.
- Reproducing and documenting software defects through structured bug reports and test plans, working directly with developers to resolve issues across sprint cycles.

Research Assistant February 2026 – Present
UNSW School of Optometry and Vision Science

- Designing a deep learning system for early glaucoma detection by applying ResNet50 and Vision Transformer architectures on multi-modal medical imaging (fundus, OCT, and visual field data).
- Experimenting with late-fusion strategies and attention-based feature extraction to combine structural data and functional data, aiming to catch pre-perimetric symptoms.
- Optimised diagnostic accuracy from an 87% baseline to 92% through hyperparameter tuning, custom loss function adjustments, and attention-based feature extraction.

Software Engineer Intern January 2026 – February 2026
PCL Construction, Sydney

- Led digital transformation of manual civil engineering workflows for large-scale solar projects by designing a full-stack automation platform, accelerating design validation phase and enhancing project delivery.
- Designed an efficient algorithmic solution, optimising operational performance by 20% and reducing manual data analysis time by 98%, significantly lowering human-error risk.
- Partnered with civil and design engineers to translate complex, deterministic logic and strict rule sets into scalable, automated data processing workflows.
- Constructed comprehensive test suites and model validation protocols to mitigate operational risks and ensure seamless deployment of enterprise-grade analytical software.

PROJECTS

Supply Chain Risk Prediction & Event Intelligence Platform February 2026 – May 2026
UNSW

- Architected an enterprise-grade risk intelligence platform ingesting large-scale meteorological and geopolitical data streams across global supply chain hubs worldwide.
- Deployed a serverless event-driven microservices architecture on AWS (Lambda, S3, DynamoDB) with automated CI/CD pipelines across dev, staging, and production environments.
- Engineered a risk-weighted pathfinding algorithm and on-demand risk analysis for any location on earth, enabling dynamic route optimisation across global supply chains. [GitHub: [Link](#)]

Invoice and Dispatch Advice API Development UNSW

April 2025 – July 2025

- Shortlisted for Macquarie Industry Prize for building a serverless invoice and dispatch advice generation platform, reducing data processing time by 40% compared to manual workflows.
- Developed a FastAPI backend on AWS Lambda to automate shipment documentation and logistics tracking.
- Designed and deployed interactive data visualisation dashboards, delivering regional logistics analytics, advanced filtering, and real-time shipment status updates to stakeholders. [GitHub: [Link](#)]

EXTRACURRICULAR ACTIVITIES

Professional Development Program Participant UNSW School of Optometry and Vision Science

October 2025 – December 2025

- Standardised Visual Field Test (VFT) and OCT imaging data using ZEISS platforms to support glaucoma detection research.
- Led data governance and lifecycle management initiatives, extracting unstructured clinical reports and transforming them into high-quality datasets to support machine learning applications.

Project Member

December 2023 – May 2024

University of New Brunswick, Canada – Manipal-GOK BioIncubator

- Selected as 1 of 20 participants from a pool of 600+ applicants for a 6-month United Nations SDG-focused entrepreneurial program.
- Conceptualised the “DigiHealth Kiosk,” a standalone healthcare solution designed to increase medical access in remote and underserved regions.
- Collaborated with a cross-border team of 4 to conduct market analysis and deliver a vetted architectural plan ready for implementation.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, R, SQL, Bash, Java

Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB, S3), Azure, Docker, Terraform

Databases: PostgreSQL, DynamoDB, MongoDB

Data Analytics & Visualisation: Tableau, Power BI

Full-Stack Development: React.js, Next.js, Flask, FastAPI

Frameworks & Libraries: PyTorch, OpenCV, TensorFlow, Scikit-learn

SOCIETIES & VOLUNTEERING

Education Subcommittee UNSW AI Society

September 2025 – Present

Sponsorship Executive UNSW Anjali Tamil Society

May 2025 – December 2025

Event Coordination Executive E-CELL, The Entrepreneurial club of MIT Manipal

January 2023 – May 2024

Creative Affairs Associate Enigma Technical Society, Manipal

September 2023 – May 2024

Volunteer Team Lead, VSO (Volunteer Service Organisation) Led community beach cleanups and youth mentorship programs.

April 2023 – May 2024

Volunteer, Kootanchoru NGO Distributed food to 1,300+ children during the COVID-19 pandemic

April 2020 – May 2021